Olympiad angle chasing

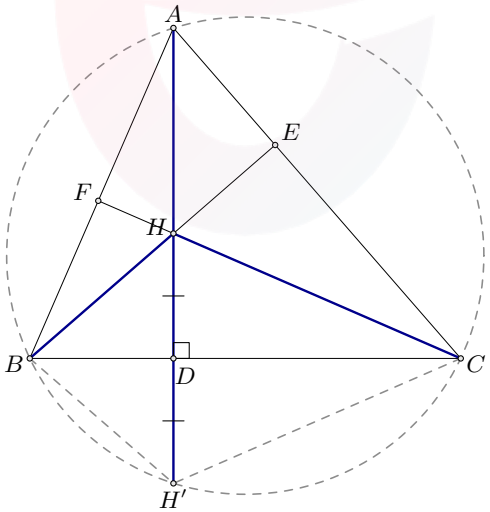
L4-GE1 Level 4 Olympiad · Geometry

Kevin Hu
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Cyclic quadrilaterals, directed angles and the standard configurations, used to write complete synthetic proofs.
By the end you can prove and use the converses of the cyclic and tangent criteria; run an angle chase with directed angles that works in every configuration; recognise orthocentre, antiparallel and incentre-excentre configurations and write BMO-standard proofs.
Prerequisites: Angle chasing and polygons (L2-GE1), Power of a point and cyclic quadrilaterals (L3-GE1). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

In an acute triangle ABC with orthocentre H, the reflection of H in BC lies on the circumcircle. To prove this, let BE and CF be altitudes. The right angles at E and F put A, F, H, E on a circle (the circle with diameter AH), so $\angle BHC = \angle FHE = 180^\circ - \angle A$. The reflection H' satisfies $\angle BH'C = \angle BHC = 180^\circ - \angle A$ and lies on the opposite side of BC from A, so the opposite angles of $ABH'C$ add to 180°, so $ABH'C$ is cyclic. This lemma is worth remembering, and we use the lemma again in Example 4.



The proof used right angles to get a circle, read an angle off that circle, and used the angle to get a second circle. Most olympiad angle chases alternate in this way between angle equalities and concyclicities. This unit covers the converses that make each step valid, directed angles to handle every configuration with one argument, the standard configurations, and what a marker expects in a written proof.

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1.1 Criteria and their converses

Cyclic criteria.
Four points A, B, C, D, no three collinear, are concyclic if and only if either of the following holds. Either $\angle ABD = \angle ACD$ with B, C on the same side of AD, or $\angle ABC + \angle ADC = 180^\circ$ with B, D on opposite sides of AC.
Tangent criterion. A line t through A touches circle ABC if and only if the angle between t and chord AB equals $\angle ACB$, the angle in the alternate segment.
Right angles. $\angle APB = 90^\circ$ if and only if P lies on the circle with diameter AB.

The forward directions are fairly trivial (the inscribed angle theorem). The method relies on the converses, and they follow from one observation. Fix the segment AD and a circle Γ through A and D. If P is inside Γ (on a given side of AD), extend AP to meet the arc at P'. Then $\angle APD$ is an exterior angle of triangle $PP'D$, so $\angle APD > \angle AP'D$. A point outside sees AD at a smaller angle in the same way. So on one side of AD, the points seeing AD at the angle given by the arc are exactly the points of that arc.

1.2 Directed angles

The weakness of these criteria is the side conditions. In a real problem a point may sit on a segment or beyond the segment, and the chase that says $\angle ABC + \angle ADC = 180^\circ$ in one picture says $\angle ABC = \angle ADC$ in another. We avoid this by measuring angles between lines, modulo 180°.

Directed angles.
For lines ℓ, m, let $\measuredangle(\ell, m)$ be the angle through which ℓ must be turned anticlockwise to become parallel to m, taken modulo 180°, and write $\measuredangle ABC = \measuredangle(BA, BC)$. Then $\measuredangle(\ell, m) = -\measuredangle(m, \ell)$, $\measuredangle(\ell, m) + \measuredangle(m, n) = \measuredangle(\ell, n)$, A, B, C are collinear iff $\measuredangle ABC = 0$, and $\measuredangle ABC + \measuredangle BCA + \measuredangle CAB = 0$ in every triangle.
Cyclic criterion. A, B, C, D are concyclic iff $\measuredangle ABC = \measuredangle ADC$. **Tangent criterion.** t touches circle ABC at A iff $\measuredangle(t, AB) = \measuredangle ACB$.

The directed cyclic criterion covers both cases. If B and D are on the same side of AC, triangles ABC and ADC have the same orientation, the directed angles are the ordinary ones with the same sign, and we recover equal angles. If they are on opposite sides, the orientations differ, say $\measuredangle ABC = \angle ABC$ and $\measuredangle ADC = -\angle ADC$, and equality modulo 180° means $\angle ABC + \angle ADC = 180^\circ$.

Two warnings. First, you cannot halve, because $2x = 2y$ only gives $x = y$ or $x = y + 90^\circ$, so central angles and angle bisectors need care. Route through a tangent instead. Second, directed angles forget which side of a line a point is on, so they cannot prove betweenness or compute "the angle is 40°". Use them for concyclicity, collinearity, parallels and tangency.

1.3 Standard configurations

The orthocentre. With feet D, E, F of the altitudes, the quadrilaterals $AFHE$, $BDHF$, $CEHD$ (diameters AH, BH, CH) and $BCEF$, $CAFD$, $ABDE$ (diameters BC, CA, AB) are all cyclic. Also $\angle BAH = 90^\circ - \angle B = \angle OAC$, so AH and AO are reflections of each other in the bisector of angle A.

Antiparallels. If X on line AB and Y on line AC make B, C, Y, X concyclic, then $\measuredangle(XY, AB) =$

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$\measuredangle YXB = \measuredangle YCB = \measuredangle ACB$, which is the angle the tangent at A makes with AB. So XY is parallel to that tangent, hence perpendicular to AO.

Incentre-excentre lemma.
Let line AI meet the circumcircle again at M, where I is the incentre and I_A the A-excentre. Then $MB = MC = MI = MI_A$.

Proof. M is the midpoint of arc BC, so $MB = MC$ and $\angle MBC = \angle MAC = \frac{1}{2}\angle A$. Hence $\angle MBI = \frac{1}{2}\angle A + \frac{1}{2}\angle B$, and $\angle MIB$, an exterior angle of triangle ABI, is also $\frac{1}{2}\angle A + \frac{1}{2}\angle B$. So $MB = MI$. The internal and external bisectors at B are perpendicular, so triangle IBI_A has a right angle at B, and the only point of the hypotenuse II_A equidistant from B and I is its midpoint, so M is that midpoint and $MI_A = MI$.

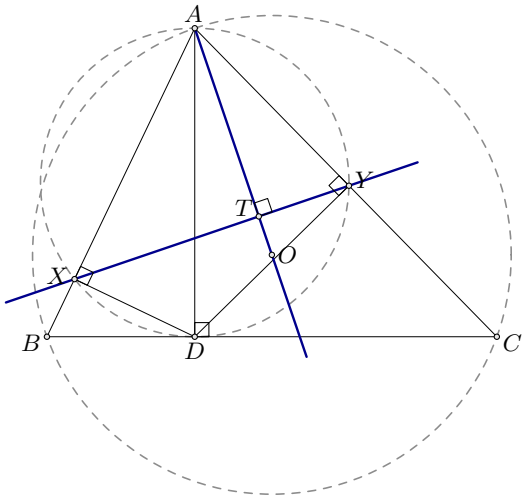
1.4 Worked examples

Example 1.
Let ABC be an acute triangle with circumcentre O, and let D be the foot of the altitude from A. Let X and Y be the feet of the perpendiculars from D to AB and AC. Prove that $XY \perp AO$.

Solution.
One could compute lengths such as $AX = AD^2/AB$, but an angle chase is shorter. Since $\angle AXD = \angle AYD = 90^\circ$, the points X and Y lie on the circle with diameter AD, so

$$\angle AYX = \angle ADX = 90^\circ - \angle DAB = \angle ABC.$$

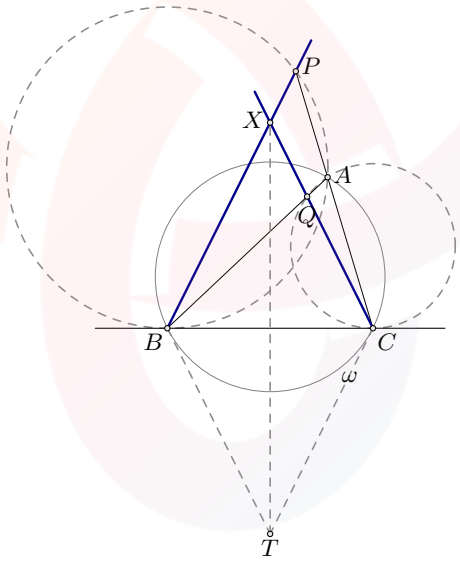
So B, C, Y, X are concyclic (exterior angle equals interior opposite angle), and by the antiparallel fact $XY \perp AO$. To finish by hand instead, note that triangle AOC is isosceles with apex angle $2\angle B$, so $\angle OAC = 90^\circ - \angle B$. If AO meets XY at T, triangle ATY has angles $90^\circ - \angle B$ at A and $\angle B$ at Y, leaving 90° at T.
Note that two right angles subtending a common segment give a circle on that segment as diameter (here the circle with diameter AD), and angles can then be transferred through that circle. You should look for such pairs before anything else.



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Example 2.
BC is a fixed chord of a circle ω, not a diameter, and A moves on ω. The circle through A tangent to line BC at B meets line AC again at P, and the circle through A tangent to BC at C meets line AB again at Q. Prove that lines BP and CQ pass through a fixed point.

Solution.
As A moves over both arcs, P can lie on segment AC, beyond A or beyond C, so an undirected chase would need cases. With directed angles one argument covers all of them.
By the tangent criterion in circle ABP, $\measuredangle(BC, BP) = \measuredangle BAP = \measuredangle BAC$, the last step because P lies on line AC. (Fixed angles for a moving point.) We look for an angle that does not change as A moves. By the cyclic criterion, $\measuredangle BAC$ is the same for every point A of ω, whichever arc A is on. Call this angle θ. So BP is the line through B at directed angle θ to BC, which is fixed. In the same way $\measuredangle(CB, CQ) = \measuredangle CAB = -\theta$, so CQ is also a fixed line.
Two fixed lines meet at a fixed point unless they are parallel, which needs $2\theta = 0$, that is $\angle BAC = 90^\circ$ and BC a diameter, which is excluded. The point can be identified. The tangent to ω at B makes angle $\measuredangle(BC, t_B) = -\theta$, so line BP is its mirror image in BC, and likewise for C. Hence the fixed point is the reflection in BC of the point T where the tangents at B and C meet.

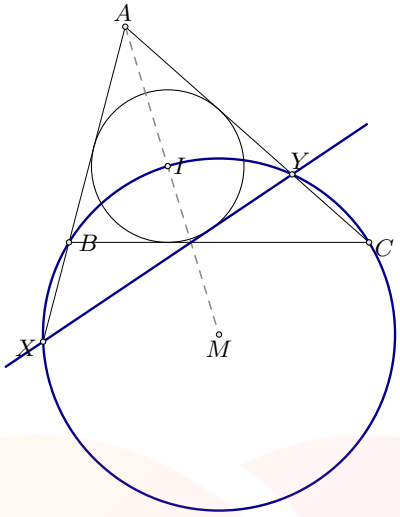


Example 3.
Let ABC be a triangle with $AB \neq AC$ and incentre I. The circle through B, I, C meets line AB again at X and line AC again at Y. Prove that XY is tangent to the incircle.

Solution.
Chasing gives $\angle AXY = \angle ACB$ quickly, since B, C, Y, X are concyclic, but tangency is a statement about the distance from I to XY, which angles at X and Y do not determine, and this is the hard step. Instead we use the incentre-excentre lemma, which says that circle BIC has centre M on line AI. A circle centred on a line is symmetric in that line, and reflection in AI swaps lines AB and AC.
So the reflection C^* of C in AI lies on the circle and on line AB, hence is B or X. The point C^* is not B, because $AC^* = AC \neq AB$. Thus $X = C^*$ and similarly $Y = B^*$, so line XY is the mirror image of line BC. The incircle is its own mirror image and touches BC, so the incircle touches XY. The argument also shows $AX = AC$ and $AY = AB$.

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(Reflection in an axis of symmetry.) When a circle's centre lies on an axis of symmetry of the figure, reflect.



Example 4.
Let ABC be an acute triangle with circumcentre O. The circle through B, O, C meets lines AB and AC again at D and E, different from B and C. Prove that the reflection of O in DE lies on the circle through A, D, E.

Solution.
The conclusion resembles the fact proved at the start of the unit, so we guess that O is the orthocentre of ADE, and aim for $DO \perp AE$. Depending on the shape of ABC, D may lie on segment AB or beyond B, so we use directed angles.
Computing $\measuredangle OCB$ is the hard step. The familiar value $90^\circ - \angle A$ comes from halving $\angle BOC = 2\angle A$, which directed angles do not allow. Use the tangent t at C instead. Since $t \perp CO$,

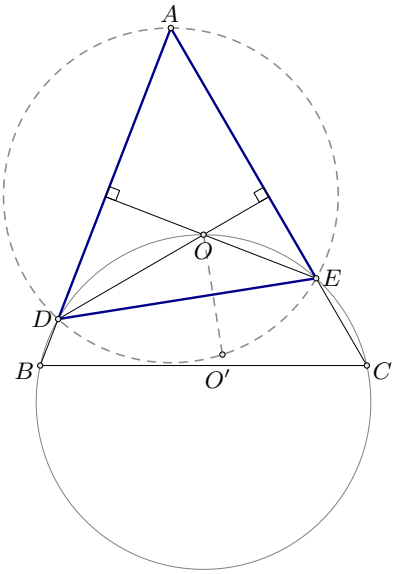
$$\measuredangle(CO, CB) = \measuredangle(CO, t) + \measuredangle(t, CB) = 90^\circ + \measuredangle CAB.$$

Now circle $BOCD$ and the line ABD give

$$\measuredangle(DO, AC) = \measuredangle ODB + \measuredangle(AB, AC) = \measuredangle OCB + \measuredangle BAC = 90^\circ + \measuredangle CAB + \measuredangle BAC = 90^\circ.$$

So $DO \perp AE$, and swapping the roles of B and C gives $EO \perp AD$, so O is the orthocentre of ADE. (The triangle is non-degenerate, since A cannot lie on circle BOC, as A sees BC at $\angle A$, not $2\angle A$.) To finish without asking whether ADE is acute, redo the opening proof with directed angles. Turning both lines through 90° gives $\measuredangle DOE = \measuredangle(AE, AD) = \measuredangle EAD$. Reflection reverses directed angles, so the reflection O' satisfies $\measuredangle DO'E = \measuredangle DAE$, and A, D, E, O' are concyclic.

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1.5 Writing up the solution

A BMO marker wants a proof that is correct in every configuration and checkable line by line. Say at the start if you use directed angles modulo 180°. Give a reason for every equality (which circle, which collinearity, which tangent). Prove a concyclicity before you use the circle. Deal with degenerate cases in a sentence. A diagram is a guide, never a justification.

Common traps.

- Proving the result only for your drawing when a point could lie on a segment or its extension.
- Halving directed angles, for instance with central angles or bisectors.
- Using "the points look cyclic" in place of a criterion.
- Circular reasoning. When the claim is that P lies on a circle, do not draw P on the circle and read off angles. Prove an angle equality, or define P' on the circle and show $P' = P$.
- Trying to get betweenness or an actual angle size out of directed angles.

1.6 How to spot this in a contest

Use these methods when the conclusion is concyclicity, collinearity, parallel or perpendicular lines, or tangency, and the data are angles, right angles, tangents or circles. I would work in the following order.

1. Mark every right angle and every tangent, since each gives a circle or an angle equality.
2. If you see an orthocentre, an incentre, an arc midpoint or a circumcentre, recall the standard facts above before chasing.
3. If the chase stalls, draw an accurate figure, look for four points that seem concyclic but have not been joined yet, and prove that as a lemma (an auxiliary circle).
4. If a point moves, ask which directed angle stays fixed.

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2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ In triangle ABC, $\angle BAC = 58^\circ$ and $\angle ABC = 80^\circ$. Let O be its circumcentre and H its orthocentre. Find $\angle OAH$.

Problem 2 ⋆⋆ In triangle ABC, $\angle A = 64^\circ$ and $\angle B = 76^\circ$. Let BE and CF be altitudes and let M be the midpoint of BC. Find $\angle EMF$.

Problem 3 ⋆⋆ Let $ABCD$ be a cyclic quadrilateral with $AB = AD$, and let its diagonals meet at E. Prove that AB is tangent to the circle through B, C and E.

Problem 4 ⋆⋆⋆ In triangle ABC, the bisector of angle A meets the circumcircle again at M, and I is the incentre. Given that $BC = 24$, $MI = 13$ and I is the midpoint of AM, find the circumradius and the perimeter of ABC.

Problem 5 ⋆⋆⋆ Let ABC be an acute triangle with orthocentre H. The circle through B, H, C meets lines AB and AC again at points D and E, different from B and C. Prove that H is the circumcentre of triangle ADE.

Problem 6 ⋆⋆⋆⋆ Let ABC be an acute triangle with circumcircle ω and circumcentre O. The tangents to ω at B and C meet at T. The line through T parallel to AB meets line AC at P. Prove that $PA = PB$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO1 2013/14 Problem 2** ⋆⋆
   Two right angles on a common segment give a circle, then bring in the tangent.
2. **BMO1 2015/16 Problem 2** ⋆⋆
   Alternate segment theorem, then an exterior angle of the cyclic quadrilateral.
3. **BMO1 2010/11 Problem 3** ⋆⋆
   Show that two angles at L add up to 180 degrees, reading each off a circle.
4. **AIME I 2003 Problem 10** ⋆⋆
   Angle chasing alone does not finish, so add an auxiliary point that creates an isosceles or equilateral triangle.
5. **BMO1 2019/20 Problem 3** ⋆⋆
   Draw the common tangent at the point of contact and chase through the tangent.
6. **BMO1 2023/24 Problem 3** ⋆⋆⋆
   Isosceles triangles at B plus inscribed angles, and directed angles avoid case splits.
7. **AIME I 2016 Problem 6** ⋆⋆⋆
   Incentre-excentre lemma, then similar triangles.

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8. **BMO1 2014/15 Problem 5** ⋆⋆⋆
   Arc midpoints give equal angles, so look for a cyclic quadrilateral through P and Q.

9. **BMO1 2015/16 Problem 5** ⋆⋆⋆
   Many right angles mean many circles, so prove collinearity by angles.

10. **USAJMO 2011 Problem 5** ⋆⋆⋆
    Bring in the centre of the circle and find a circle through P and the tangency points.

11. **USAJMO 2016 Problem 1** ⋆⋆⋆⋆
    Guess the fixed point from a special position of P, and arc midpoints are centres of circles through incentres.

12. **USAJMO 2018 Problem 3** ⋆⋆⋆⋆
    Reflections make B equidistant from three points, so check whether Q lies on a line through B.

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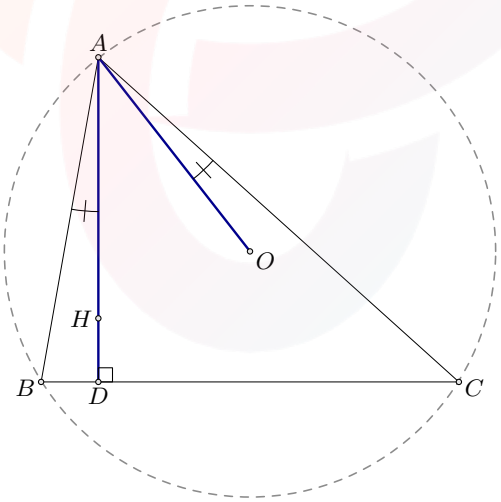
4 Hints

1. Work out $\angle BAH$ and $\angle OAC$ separately in terms of the angles of the triangle.
2. Where is the centre of the circle through B, C, E, F?
3. Use the converse of the alternate segment theorem and compare the angle between AB and BE with $\angle BCE$.
4. The incentre-excentre lemma tells you MB. Then look at the right-angled triangle formed by M, the midpoint of BC and B.
5. Prove $HA = HD$ by showing the base angles of triangle HAD are equal. D may be on either side of B, so use directed angles.
6. First show that P lies on the circle through T, B, C. Which other notable point lies on that circle, and what does that say about the angle at P?

5 Solutions

Problem 1

In triangle ABC, $\angle BAC = 58°$ and $\angle ABC = 80°$. Let O be its circumcentre and H its orthocentre. Find $\angle OAH$.



The triangle has $\angle C = 42°$, so the triangle is acute and both O and H lie inside the triangle. Since $AH \perp BC$, the right-angled triangle ABD (with D the foot of the altitude) gives $\angle BAH = 90° - \angle B = 10°$. Triangle AOC is isosceles with $\angle AOC = 2\angle B = 160°$, so $\angle OAC = 10°$ as well.

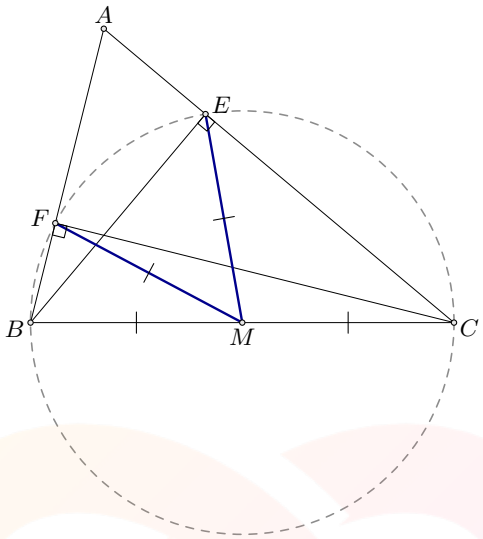
The rays AH and AO therefore make $10°$ with AB and with AC respectively, and since $10° + 10° < 58°$ they lie in that order inside angle BAC. Hence $\angle OAH = 58° - 10° - 10° = 38°$. In general $\angle OAH = |\angle B - \angle C|$, which is worth remembering.

Answer: $38°$

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Problem 2

In triangle ABC, $\angle A = 64^\circ$ and $\angle B = 76^\circ$. Let BE and CF be altitudes and let M be the midpoint of BC. Find $\angle EMF$.



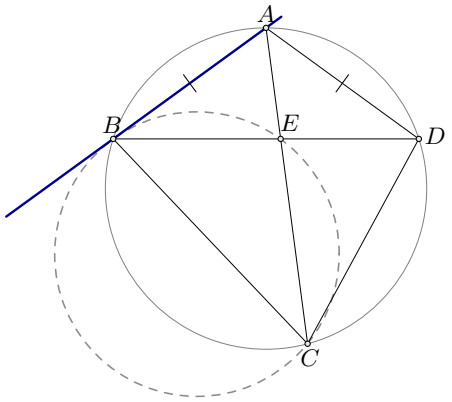
Here $\angle C = 40^\circ$, so the triangle is acute and E, F lie on the sides. The right angles $\angle BEC = \angle BFC = 90^\circ$ put E and F on the circle with diameter BC, whose centre is M. So $MB = MC = ME = MF$.

Triangle MEC is isosceles, so $\angle MEC = \angle MCE = 40^\circ$ and $\angle EMC = 100^\circ$. Triangle MFB is isosceles, so $\angle MFB = \angle MBF = 76^\circ$ and $\angle BMF = 28^\circ$. These two angles sit on either side of $\angle EMF$ along the line BC, so $\angle EMF = 180^\circ - 100^\circ - 28^\circ = 52^\circ$. Equivalently, the angle at the centre is twice $\angle ECF = 90^\circ - \angle A = 26^\circ$.

Answer: 52°

Problem 3

Let $ABCD$ be a cyclic quadrilateral with $AB = AD$, and let its diagonals meet at E. Prove that AB is tangent to the circle through B, C and E.



Since $ABCD$ is a convex cyclic quadrilateral, the diagonal BD separates A from C, and E lies on

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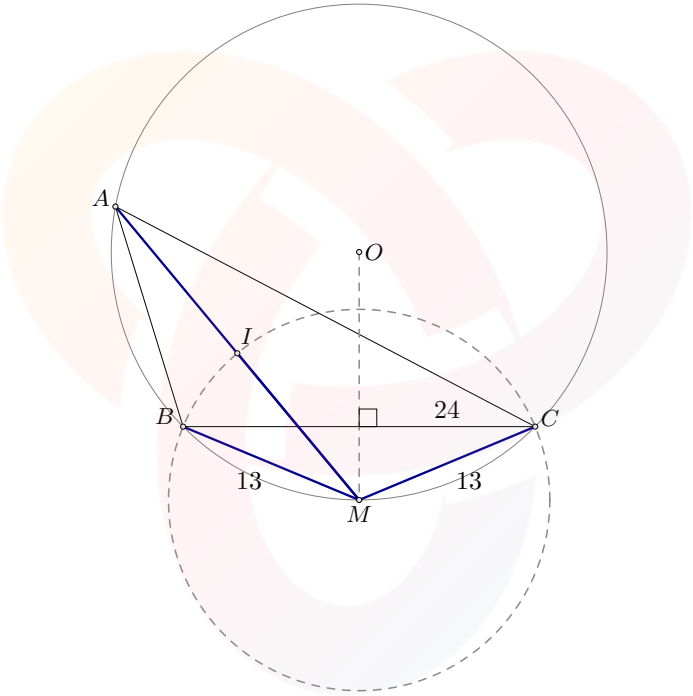
both diagonals. Because $AB = AD$, triangle ABD is isosceles and $\angle ABD = \angle ADB$. Angles subtending the same arc AB are equal, so $\angle ADB = \angle ACB$.

Hence $\angle ABE = \angle ABD = \angle ADB = \angle ACB = \angle ECB$, using that E lies on segments BD and AC. So the angle between line AB and the chord BE of circle BCE equals the angle $\angle BCE$ subtended by BE at C, and A and C lie on opposite sides of BE. By the converse of the alternate segment theorem, AB is tangent to circle BCE at B. (With directed angles, $\measuredangle(BA, BE) = \measuredangle ABD = \measuredangle BDA = \measuredangle BCA = \measuredangle BCE$, which is the tangent criterion.)

Note that the power of A with respect to circle BCE then gives $AB^2 = AE \cdot AC$.

Problem 4

In triangle ABC, the bisector of angle A meets the circumcircle again at M, and I is the incentre. Given that $BC = 24$, $MI = 13$ and I is the midpoint of AM, find the circumradius and the perimeter of ABC.



By the incentre-excentre lemma, $MB = MC = MI = 13$. As M is the midpoint of arc BC, the line OM is the perpendicular bisector of BC. Let this line meet BC at N. Then $BN = 12$ and, from the right angle at N, $MN = \sqrt{13^2 - 12^2} = 5$.

The centre O lies on line MN with $OM = R$, so $ON = |R - 5|$ and Pythagoras in triangle OBN gives $12^2 + (R-5)^2 = R^2$, that is $169 = 10R$, so $R = \frac{169}{10}$.

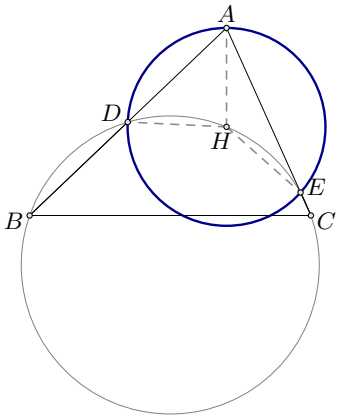
For the perimeter, $AM = 2 \cdot 13 = 26$ (which is less than the diameter 33.8, so such a triangle exists). Ptolemy's theorem in the cyclic quadrilateral $ABMC$ gives $AB \cdot MC + AC \cdot MB = AM \cdot BC$, so $13(AB + AC) = 26 \cdot 24$ and $AB + AC = 48$. The perimeter is $48 + 24 = 72$.

Answer: $R = \frac{169}{10}$, perimeter 72

Problem 5

Let ABC be an acute triangle with orthocentre H. The circle through B, H, C meets lines AB and AC again at points D and E, different from B and C. Prove that H is the circumcentre of triangle ADE.

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We use directed angles modulo 180°, since D may lie on segment AB or beyond B (and similarly for E). We show $HA = HD$, and the same argument with B and C swapped gives $HA = HE$.

First, triangle HDA is non-degenerate. $D \neq A$, because A is not on circle BHC, because A and H are on the same side of BC and $\angle BHC = 180° - \angle A \neq \angle A$ as the triangle is acute. Also H is not on line AB, since the triangle is acute.

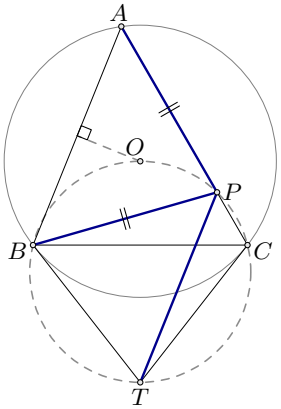
Using that A, B, D are collinear and then that B, H, C, D are concyclic, $\measuredangle HDA = \measuredangle HDB = \measuredangle HCB$. As $CH \perp AB$, $\measuredangle HCB = \measuredangle(CH, AB) + \measuredangle(AB, BC) = 90° + \measuredangle(AB, BC)$. On the other hand, as $AH \perp BC$, $\measuredangle DAH = \measuredangle(AB, AH) = \measuredangle(AB, BC) + \measuredangle(BC, AH) = \measuredangle(AB, BC) + 90°$.

So $\measuredangle HDA = \measuredangle DAH$. In the triangle HDA these two directed angles are the angles at D and A, both measured in the sense of the triangle's orientation, so they equal the ordinary angles with the same sign. Two angles of a triangle that agree modulo 180° are equal, so $\angle HDA = \angle DAH$ and $HA = HD$. Together with $HA = HE$, H is equidistant from A, D, E, so H is the circumcentre of ADE.

Compare Example 4, where the circle through B, O, C produced a triangle with orthocentre O.

Problem 6

Let ABC be an acute triangle with circumcircle ω and circumcentre O. The tangents to ω at B and C meet at T. The line through T parallel to AB meets line AC at P. Prove that $PA = PB$.



We use directed angles modulo 180°. If $CA = CB$ then the tangent at C is parallel to AB, so $P = C$ and there is nothing to prove. Assume $CA \neq CB$, so $P \neq C$. Also $P \neq T$, since T is not on line AC (otherwise AC would be the tangent at C).

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Step 1. T, P, B, C are concyclic. As $PT \parallel AB$ and P lies on line AC, $\measuredangle TPC = \measuredangle(AB, AC) = \measuredangle BAC$. By the tangent criterion for the tangent TB, $\measuredangle TBC = \measuredangle(BT, BC) = \measuredangle BAC$. Hence $\measuredangle TPC = \measuredangle TBC$, and T, P, B, C are concyclic.

Step 2. O lies on this circle too. The radii OB, OC are perpendicular to the tangents, so $\angle OBT = \angle OCT = 90^\circ$ and B, C lie on the circle with diameter OT. That circle is the circle TBC, so P lies on the circle with diameter OT. If $P = O$ then $PA = PB$ trivially. Otherwise $\angle OPT = 90^\circ$.

Step 3. So $OP \perp PT$, and $PT \parallel AB$, hence $OP \perp AB$. The line through O perpendicular to AB is the perpendicular bisector of AB, so P lies on this line and $PA = PB$. ■

The same argument shows that P lies on circle BOC, which links this problem to Example 4.